

Problem

A 240/2,400-V rms step-up ideal transformer delivers 50 kW to a resistive load. Calculate:

(a) the turns ratio

(b) the primary current

(c) the secondary current

Step-by-step solution

Step 1 of 3

(a)

This is a step-up transformer, since $V_1 = 240 \text{ V} < V_2 = 2400 \text{ V}$

Determine the turns ratio.

$$n = \frac{V_2}{V_1}$$

Substitute 2400 V for V_2 , 240 V for V_1 .

$$\begin{aligned} n &= \frac{2400}{240} \\ &= 10 \end{aligned}$$

Therefore, the turns ratio is 10.

Step 2 of 3

(b)

Output power for ideal transformer,

$$P = V_2 I_2 = V_1 I_1$$

Substitute 50 kW for P , 240 V for V_1 .

$$\begin{aligned} I_1 &= \frac{P}{V_1} \\ &= \frac{50 \times 1000}{240} \\ &= 208.3 \text{ A} \end{aligned}$$

Therefore, the primary current is 208.3 A.

Step 3 of 3

(c)

At transformer terminals,

$$n = \frac{I_1}{I_2}$$

Substitute 208.3 A for I_1 and 10 for n .

$$\begin{aligned} I_2 &= \frac{208.3}{10} \\ &= 20.8 \text{ A} \end{aligned}$$

Therefore, the secondary current is 20.83 A.